

F(AI)²R:

FAIR Research with AI in the Loop, Twice

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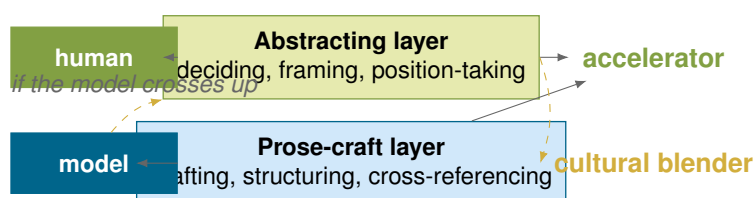
Abstract

The FAIR Guiding Principles — Findable, Accessible, Interoperable, Reusable — were drafted for data and have since been extended to research software (FAIR4RS) and machine-learning models (FAIR4ML). Neither extension yet covers the artefact classes that Large-Language-Model-assisted scientific work has produced: exportable conversation transcripts, versioned prompt files, tool-and-model manifests, verification-status ladders, structured redaction policies, and per-claim provenance maps. We propose $F(AI)^2R$, read *F-A-I-A-I-R*: the canonical FAIR axes with an additional (*AI*) factor multiplied through them, squared because each axis demands two passes over every artefact — authoring and audit. The contribution is not the name; it is the integration of eight individually unoriginal practices into a single discipline, enforced by a ten-stage agent pipeline whose prose-owning and audit-owning roles are strictly separated through a handback discipline. We release this paper, its prompts, and its PROV-O graph as the worked example.

Author’s Note — Advice for reading

A short note from me (Florian Krebs), placed up front because *how* you read this paper matters as much as *what* is in it. This is a position paper, not a survey: the contribution is a stance, that FAIR is insufficient for LLM-assisted scholarly writing in its current form and that the eight integrated practices of §4 are the minimum discipline an honest author can adopt today. The paper defends that stance; a reader who comes looking for a survey will find an argument. $F(AI)^2R$ is a snapshot of a moving target — model capabilities, disclosure norms, and licensing postures will move under us; the version manifest in `doc/provenance.ttl` is the receipt that lets a future reader date what we committed to.

The partition the cooperation makes visible. A second-order observation, recorded for the recursive case study (§6). Cooperative writing with an LLM, as practised here, partitions cleanly: the human does the abstraction, the logical reasoning, and the choice of the next step; the model takes those decisions and turns them into readable prose. *The human does the deciding, the model does the drafting, and naming the partition is itself a methodological act.* The pattern this paper describes is designed to make that partition legible from outside the session — in the prompts, in the transcripts, and in the graph. The human-side contribution is structural and corrective (deciding the position, abstracting the eight-practice integration, catching drift, staking reputation under a published name); the model-side contribution is volume and second-order pattern-matching (drafting at scale, cross-referencing, keeping graph and manuscript in sync). Human interventions are rare in volume but high in leverage (Figure 2); model outputs are long and constant but low in irreversibility.



Diagnostic: would the abstraction have been thinkable without the model?

if yes: **accelerator**
if no: **open question**

Figure 1. The coupling rule that sharpens the human-deciding / model-drafting partition into a normative diagnostic. The cooperation works as an *accelerator* (right-top, green) when the model stays on the prose-craft side of the partition (solid arrow). It collapses toward a *cultural blender* (right-bottom, yellow) when the model is allowed across into the abstracting layer (dashed arrows). The honest diagnostic, recorded in `doc/user-observations-log.md`: *ask whether the abstraction would have been thinkable without the model*. See §.

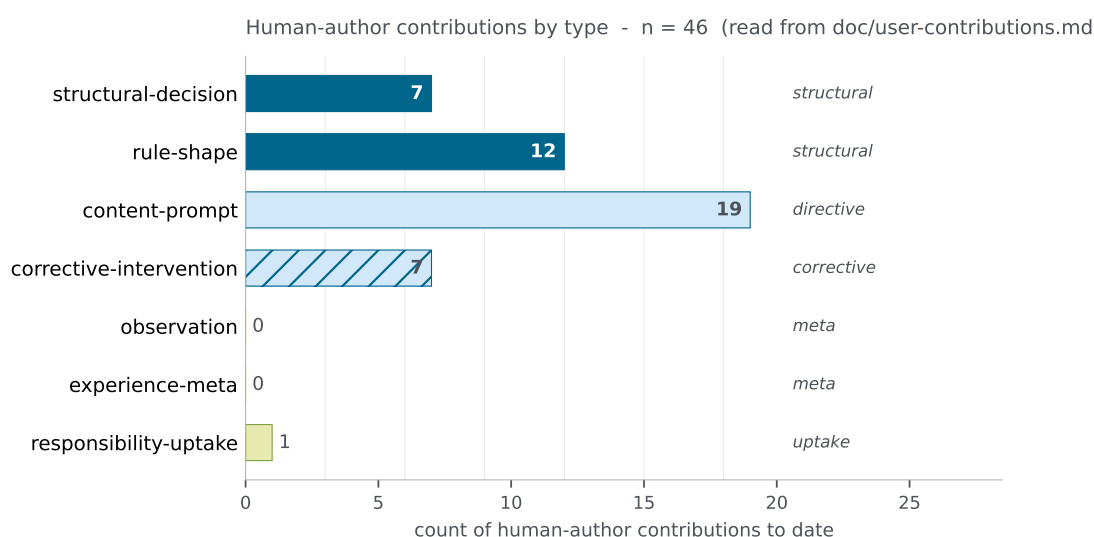


Figure 2. Human-author contributions to the manuscript by type, read from `doc/user-contributions.md`. Structural decisions and rule-shape interventions are rare but high leverage; corrective interventions are the catch-and-fix work; responsibility-uptake is the leg of the contribution the AI structurally cannot share. Regenerated on every build.

A coupling rule sharpens the partition (Figure 1). The cooperation works as an *accelerator* when the model stays on the prose-craft side of the partition — drafting, cross-referencing, structuring — where its strength is interpolation within the manifold of prior text [1, 2, 3]

verify

. It collapses toward a *cultural blender* when the model is allowed across into the abstracting layer, where its failure mode is to smooth the unfamiliar into the nearest familiar shape [4]

verify

. The diagnostic, recorded in the observations log: *ask whether the abstraction would have been thinkable without the model*. If yes, the model was an *accelerator* on that abstraction; if no, the question genuinely opens. F(AI)²R is a case where the answer is yes; the partition is what kept it that way.

Tokens out, tokens in. $\text{\LaTeX}$, prompt files, Turtle, and the build pipeline are committed under permissive licences and remain machine-readable. That is deliberate. A future agent compressing or extending this work has the inputs needed to do so without laundering its provenance through a binary PDF. The tokens go in; the tokens come back out; the chain of attribution is what makes the round-trip auditable.

1 Introduction

The FAIR Guiding Principles for scientific data management [5] were drafted for data, then extended to research software (FAIR4RS [6]) and to machine-learning models (FAIR4ML [7]). Neither extension yet covers the artefact classes that LLM-assisted scholarly writing produces: exportable conversation transcripts, versioned prompt files, tool-and-model version manifests, verification-status ladders, structured redaction policies, and per-claim provenance maps.

This is not a hypothetical gap. Anyone now writing a paper with an LLM in the room is producing those artefacts, whether or not they preserve them. If they vanish, two things follow. First, the audit trail is gone — a reader cannot ask *who wrote this sentence, what evidence supports this claim, and how was the claim verified?* [8]. Second, the work is more exposed to the documented failure modes of LLM-assisted writing, which sit on top of a reproducibility baseline that is already poor in many fields [9, 10]

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: fabricated citations, drift between claim and evidence, sloppification, and the gradual flattening of an argument as hedges accrete. Reported base rates are not anecdotal: Walters and Wilder [11] report 55% fabrication for GPT-3.5 and 18% for GPT-4 across general-purpose generation; Magesh et al. [12] report 17–34% hallucination for retrieval-augmented legal-research tools. The numbers are not directly comparable — one grades full citations, the other grades RAG-anchored answers — but they are the same order of magnitude, and it is the kind of number that turns a methodological inconvenience into a disqualifying one.

The volume problem compounds the per-paper one. The fraction of journal and arXiv submissions that contain LLM-generated or LLM-assisted prose has risen sharply since 2023 [13, 14]; detection studies report measurable increases across computer science, biomedicine, and the social sciences [15]. Editors at major venues have publicly described AI-shaped text as a load they cannot review out of submissions one paragraph at a time, and have asked for provenance *from* the authors instead of post-hoc detection *against* them [16, 17]

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. Paper mills, fabricated literatures, and the boilerplate flattening of disclosure pages are the symptoms; what is missing is a discipline that authors can adopt on day one rather than a forensic regime that reviewers apply on the back end. Detection alone is, on present evidence, an arms-race the reviewer side does not win [18]

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; an authoring-side discipline is the cheaper, sturdier complement. The pattern this paper proposes is meant to be that discipline.

The journal as distribution channel may be in decline. The strain is not only on the per-paper review pipeline; it is on the format itself. The original concept of the journal — a periodical bundling vetted articles for distribution to subscribers — pre-dates the internet by three centuries, and the network has, on aggregate, weakened the case for bundling. Preprint servers, overlay journals, registered reports, open-science platforms, and artefact-versioned repositories now cover much of what journals historically did, often faster and more openly [19, 20]

verify

. The arrival of LLM-generated submissions on top of a system that was already showing format strain is what tips a slow drift into a visible crisis: editors face submission queues swollen by AI-fabricated content [21]

verify

at the same time that authors face venue-policy churn [22, 23, 24] and reviewers face capacity exhaustion. We do not claim the journal is dead — the position would be over-strong, and many fields still need the curation work journals do. We do claim that the assumption that the journal is the unproblematic default for scholarly distribution can no longer be taken for granted, and that an authoring-side discipline whose primary outputs are the *repository*, the *provenance graph*, and the *manuscript bundled together* is at least as well-suited to the emerging communication ecosystem as it is to the legacy one. F(AI)²R is designed to fit both.

Since Vaswani et al. [25] put scaled dot-product attention at the centre of sequence modelling, the population of plausible authors of a scientific paragraph has changed in kind, not just in degree; FAIR's 2016 vintage predates that shift by a year, and its operational vocabulary still assumes that the agent at the keyboard is a person.

We propose $F(AI)^2R$, read *F-A-I-A-I-R*: the canonical FAIR axes with an additional (*AI*) factor multiplied through them, squared because each axis demands two passes over every artefact, an authoring pass and an audit pass. An earlier draft used *FAIR4AI*; we discarded it after observing that *4AI heißt alles* — a suffix denotes nothing about how the principles change. Folding (*AI*) into the acronym, multiplied through, asserts the extension is internal to FAIR rather than external to it.

Contribution. The contribution is not the name. It is the integration of eight individually unoriginal practices into a single discipline: (1) transcript preservation; (2) verification-status labelling; (3) per-claim provenance maps; (4) mirror discipline between source and render; (5) recursive meta-process documentation; (6) base-rate-anchored AI disclosure; (7) legal honesty about authorship; and (8) FAIR alignment treated as a precondition, never as a retrofit. None of the eight is novel on its own. Their joint adoption, enforced by a ten-stage agent pipeline with strict separation between prose-owning and audit-owning agents, is what we claim is new. *The pipeline is the team; the team is the methodology.*

Roadmap. §2 states the F(AI)²R position in five numbered claims. §3 states the pattern itself: the axes, the ten-stage pipeline, the verification ladder as a finite-state machine, and the handback discipline. §4 treats the eight integrated practices. §5 reports the audit pass on the live graph; §6 narrates how the pipeline itself evolved during cooperative writing. §7 names the failure modes the pattern resists and is honest about which it does not. §8 addresses the dual-use surface, equity, and the standing objections; §9 states the conditions for community-recommendation submission. Seven appendices supply the verification-ladder specification, the prompt taxonomy, the F(AI)²R cell mapping, the process-evolution chronology, the provenance-analysis queries, the implementation notes, and the pattern's two forward-looking extensions (domain ontologies and graph-as-research-infrastructure).

2 Position

This paper takes a position. The five claims below are the position in compressed form. They are stated here, defended in §3–§4, exercised on the present manuscript in §F–§6, and tested against the failure modes a thoughtful reader will name in §8.

1. **FAIR is insufficient for LLM-assisted scholarly writing in its current form.** The fifteen

(AI)² factor multiplied through every axis

	Findable	Accessible	Interoperable	Reusable
(AI) ₁ authoring pass	DOI / ORCID; IRIs minted	clone-and-build; licences	Turtle graph; fair2r: namespace	prompts; Makefile; logbook
(AI) ₂ audit pass	metadata agree; IRIs unique	paywalled sources vendored	Turtle validates; SHACL	clone replays; drift reported

Figure 3. The F(AI)²R pattern as a 4 × 2 grid. The four FAIR axes (top, grey) each carry an authoring pass AI₁ (blue, middle row) and an audit pass AI₂ (green, bottom row); the squared notation is the typographic commitment that the (AI) factor is multiplied through every axis rather than appended as a fifth one. Cell contents summarise §3.1; the full mapping is in Appendix C.

sub-principles of Wilkinson and colleagues [5] were drafted for data, and the FAIR4RS [6] and FAIR4ML [7] extensions were drafted for software and models. None covers the artefact classes that LLM-assisted writing produces: transcripts, prompts, version manifests, verification ladders, redaction policies, per-claim provenance maps. A third extension is required.

- The required extension is multiplicative, not appended.** Folding (AI) into the FAIR acronym — $F(AI)^2R$, with the (AI) factor passed through every axis and squared because each axis demands two passes (authoring and audit) — asserts that the change is *internal* to FAIR rather than external to it. A *4AI* suffix denotes nothing about how the principles change; *4AI heit alles*.
- Eight integrated practices are the minimum discipline.** Transcript preservation, verification-status labelling, per-claim provenance maps, mirror discipline, recursive meta-process documentation, base-rate-anchored disclosure, legal honesty about authorship, and FAIR alignment as precondition. None of the eight is novel on its own. The integration is the contribution, and adopting any subset is, on the failure modes the literature has documented, insufficient.
- The pipeline is the team; the team is the methodology.** A ten-stage agent pipeline with strict handback between prose-owning and audit-owning agents (§3.2) is the cheapest known mechanism that makes the eight practices enforceable rather than aspirational. Specialisation reduces hallucination at the seam.
- Naming the practice is the first step toward contesting it.** F(AI)²R is offered as a position to be argued with — contestable in the open literature, replayable on a clean clone of the public repository, and revisable as the capability of the underlying models, the disclosure norms of the venues, and the licensing posture of the corpora that train the next models all move under us.

The remainder of the paper defends each of the five claims. A reader who disagrees with any one of them is invited to say so and to bring evidence; the artefact discipline this paper prescribes is engineered specifically so that the disagreement can be checked against the same graph the position rests on.

3 The F(AI)²R Pattern

		writer	human	AI
reader	human	classical scholarship <i>all four FAIR axes</i> prose, citations, manuscripts and prompts	LLM-assisted writing <i>the F(AI)²R primary case</i> prose + transcripts + prompts + ladder-status + per-claim provenance map.	
	AI	AI-assisted reading <i>auditor / scrutinizer</i> machine-readable claims, SPARQL-able graph ingestion.	AI-to-AI corpus <i>training data / ML retrieval / agent-mediated review</i> everything above, plus base-rate disclosure on the training-set side; collapse risk.	

disclosure surface:
*paper bundle widens
 top-left → bottom-right*

The (AI) factor squared: both writer and reader can be human or AI. The paper's primary case is the top-right cell; its audit

Figure 4. The (AI)² factor unpacked as a 2 × 2 grid of writer (columns) and reader (rows). The top-left cell (human writer, human reader) is the classical scholarly case; the top-right cell (AI writer, human reader) is this paper's primary case and is what every practice in §4 addresses; the bottom-left cell (human writer, AI reader) is the audit / scrutinizer / agent-mediated search case and requires machine-readable claims and a SPARQL-able provenance graph; the bottom-right cell (AI writer, AI reader) is the systemic case (training-data ingestion, agent-to-agent review). The disclosure surface widens monotonically along the diagonal.

3.1 Axes: F(AI)²R

We retain the four canonical FAIR axes and read each in two passes (Figure 3). The (AI)² factor itself unpacks into the four cells of writer-vs-reader (Figure 4): a manuscript that will be both written and read by humans only needs the manuscript proper, while a manuscript whose readers include AI agents needs every artefact in the F(AI)²R bundle. The authoring pass (AI₁) produces the artefact for that axis. The audit pass (AI₂) verifies the artefact against the axis after the fact, never in the same role and never in the same commit. The two passes are what justifies the squared notation. The (AI) factor is multiplied through every axis.

F. Authoring mints persistent identifiers (DOI via Zenodo, ORCID for authors), populates machine-readable metadata (CITATION.cff, codemeta.json, .zenodo.json), and assigns IRIs to per-claim entities. Audit verifies the metadata surfaces agree, no IRI is dangling, and every manuscript-cited claim has a graph entry.

A. Authoring clones-and-builds: a third party with the public repository and a documented toolchain reproduces the PDF. Audit verifies the licence pair (MIT for code/RDF, CC-BY-4.0 for prose), that paywalled sources have been vendored where licensable, and that the redaction policy covers what it claims.

I. Authoring commits the provenance graph as PROV-O [26] over Turtle in doc/provenance.ttl. Audit validates the Turtle, runs SHACL shapes if defined, and verifies every section has a prov:Activity record that produced it.

R. Authoring commits the per-role prompts under agents/, the build target under paper/Makefile, and the logbook under doc/logbook.md. Audit replays the pipeline on a clean clone and reports drift.

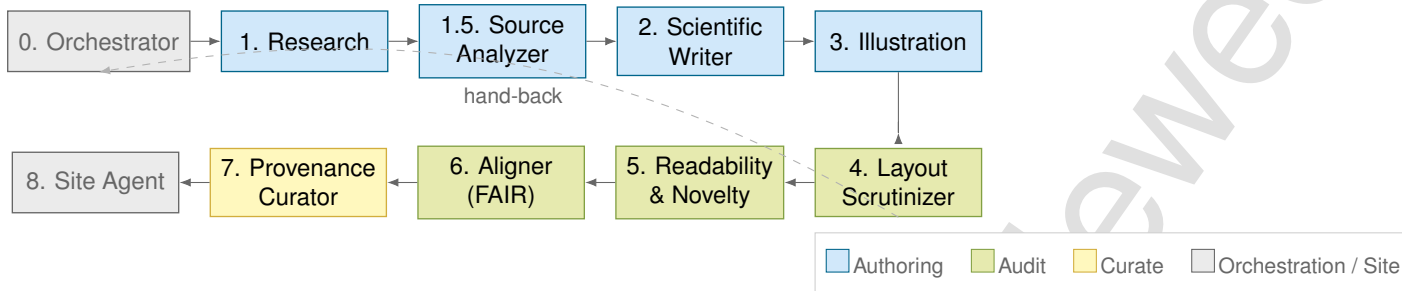


Figure 5. The ten-stage F(AI)²R pipeline. Authoring (blue) and audit (green) stages are strictly separated by the handback discipline (§3.4); only the Provenance Curator (yellow) writes `doc/provenance.ttl`; the Orchestrator and Site Agent (grey) are coordination roles that never edit prose. A scrutinizing stage that finds a defect files a *handback* (dashed arrow) rather than patching in place.

3.2 The pipeline as the team

Ten stages, coordinated by an Orchestrator, with strict handback between prose-owning and audit-owning agents (Figure 5):

- 0. Orchestrator.** Inspects state, dispatches one downstream stage at a time. Never edits prose.
- 1. Research.** Per-section literature audit. Produces sources and transcripts under `doc/sources/` and `doc/transcripts/`.
- 1.5. Source Analyzer.** Promotes LIT-RETRIEVED → AI-CONFIRMED; never crosses into LIT-READ.
- 2. Scientific Writer.** Owns prose, register, $\text{\LaTeX}$. Does not change claims or introduce citations without a verified source.
- 3. Illustration.** Owns figures; halts on load-bearing values still at LIT-RETRIEVED.
- 4. Layout Scrutinizer.** Reads the rendered PDF, files a Layout Defect Registry under `doc/handbacks/`.
- 5. Readability & Novelty Scrutinizer.** Reads the markdown, flags repetition, list-of-lists prose, hedging chains, and unsupported novelty. Files a registry.
- 6. Aligner (FAIR).** Audits traceability across paper, sources, README, and registries. Routes hand-backs; never edits content.
- 7. Provenance Curator.** Sole writer of `doc/provenance.ttl`; refuses to invent triples.
- 8. Site Agent.** Owns the public site; refuses to render without explicit publication consent. The site never introduces a new claim.

The agent that owns prose cannot mark its own claims as verified. The agent that audits layout cannot edit prose. The agent that curates provenance cannot write $\text{\LaTeX}$. *Specialisation reduces hallucination at the seam.*

3.3 The verification ladder as an FSM

Every claim carries a verification state. The states are organised as a finite-state machine (Figure 6), monotone non-decreasing modulo explicit retraction:

UNVERIFIED-EXTERNAL → NEEDS-RESEARCH → LIT-RETRIEVED → AI-CONFIRMED → LIT-READ

LIT-RETRIEVED: a candidate URL or DOI that resolves. AI-CONFIRMED: an AI agent has read the source and confirmed it supports the claim, with a quoted snippet preserved in `doc/sources.md`. LIT-READ: a human has read the source and confirmed both that it supports the claim and that



Figure 6. The verification ladder as a finite-state machine. A `fair2r:Claim` progresses left-to-right as evidence accumulates; AI-CONFIRMED is the ceiling an LLM agent can deliver on its own (§3.3). Retraction is the only back-edge and is recorded as a `prov:Invalidation` activity in `doc/provenance.ttl`, never as a deletion. A claim may be *invoked* at its current rung or below; load-bearing or contested claims still require LIT-READ. Source: Mermaid stateDiagram-v2 at `paper/figures/ladder-fsm.mmd`, mirrored on the public site.

the inference from source to claim is sound. A claim may invoke a source at its current rung or below; load-bearing or contested claims still require LIT-READ.

The AI-CONFIRMED rung exists because verification overhead is real and dominates writer cycles. Most retrieved entries are obvious from the abstract; a small minority anchor contested

claims; the single-tier gate forced the human reader through both populations at the same cost. Splitting the work between the Source Analyzer (cheap) and the human (expensive) is the lever that makes the practice economically tenable.

Crosswalk. Retrieval depth maps to PRISMA 2020 [27]: LIT-RETRIEVED = Identification, AI-CONFIRMED = Screening, LIT-READ = Included. Invocation strength maps to GRADE [28]: a claim invoked at LIT-READ carries a GRADE certainty floor of MODERATE; a claim at AI-CONFIRMED can carry at most LOW. The mapping is a floor, not a ceiling.

3.4 Handback discipline

The four scrutinizing stages — Source Analyzer, Layout Scrutinizer, Readability & Novelty Scrutinizer, Aligner — never edit source. They write *Defect Registries* under `doc/handbacks/`; the repair lands in a subsequent commit by a prose-owning agent. The cost is one extra commit boundary per cycle. The payoff is that “what did the scrutinizer actually flag?” has a single canonical answer in the git history. The mirror discipline that backs the handback — every primary artefact has a (source, render) pair under version control, the build either agrees or fails — is enumerated as practice 4 of §4.

Extensions and cousins. The pattern composes with two forward-looking extensions — domain ontologies (SOSA / SSN, OM-2 / QUDT) for sub-discipline value semantics, and provenance-graph interconnectivity (the bioinformatics precedent under data-volume pressure, restated for LLM-assisted scholarship), both recorded in Appendix G. Read structurally, F(AI)²R borrows more from formal methods than from the data-management literature. Each `fair2r:Claim` IRI is a conjecture; the verification ladder is a proof-state machine; the audit pass is a proof-checker that verifies structural properties of the graph rather than re-discovering them; the SHACL shapes (Appendix G) are invariants; the handback discipline is the prover / checker separation that model checking and certified-software efforts treat as load-bearing [29, 30]

verify

. The analogy is partial — a manuscript is not a transition system, and LIT-READ is not soundness — but the imports it suggests (counter-example witnesses on retracted claims; lemma libraries of vetted claims that successor papers can cite by IRI without re-verifying) are flagged as future work.

4 Eight Integrated Practices

Each practice below is individually unoriginal. The argument is that adopting any one in isolation is insufficient against the failure modes catalogued in §7, while adopting all eight, with the pipeline of §3.2 enforcing the seams, is sufficient against most and honest about the rest. Figure 8 reads the eight as columns against the failure modes as rows, showing which practice carries the load for each.

- 1. Transcript preservation.** Every working session that touches the manuscript exports its conversation transcript and commits it under `doc/transcripts/`. Transcripts are bound to the manuscript via `prov:used` on the corresponding authoring activity, typed as `fair2r:Transcript` in the provenance graph.
- 2. Verification-status labelling (§3.3).** Every claim carries one of the five FSM rungs. The rung is recorded in the graph, not in the prose; a claim invoked at a rung lower than its evidence permits is allowed, a claim invoked higher is a defect surfaced by the Aligner.
- 3. Per-claim provenance maps.** Every named claim has a `fair2r:Claim` IRI in `doc/provenance.ttl` with at minimum `prov:wasGeneratedBy`, `prov:wasAttributedTo`, `prov:wasDerivedFrom`, and

`fair2r:verificationState`. A retraction is a `prov:Invalidation` activity, never a deletion. The per-claim discipline is the structured-documentation descendant of Datasheets for Datasets [31] and Model Cards [32], applied at claim granularity rather than at artefact level.

4. **Mirror discipline (§3.4).** Every primary artefact has a (source, render) pair under version control. The build either agrees or fails. *Asset honesty over asset completeness*: a placeholder is allowed and encouraged, but it must visibly identify itself as AI-authored and pending.
5. **Recursive meta-process documentation (§6).** A paper that prescribes a discipline must have been written under that discipline. This paper is its own case study — the manuscript, the agent prompts, the provenance graph, the logbook, and the build pipeline are all in the public repository, and the cooperation that produced them is narrated in §6 and chronicled in Appendix D.
6. **Base-rate-anchored AI disclosure.** Disclosure is anchored to numeric base rates for the failure modes that the rungs are supposed to mitigate. For fabricated citations specifically, we cite the figures that motivated the rung's existence [11, 12] and declare the fraction of claims at each rung at submission time.
7. **Legal honesty about authorship.** AI is acknowledged but is not an author. The position holds across Germany (§2 *UrhG*), the United States, and the major medical journals [33]: AI cannot hold copyright, cannot consent to publication, cannot be held accountable. The `CITATION.cff` lists humans only; the graph attributes activities to AI agents but never promotes them to authors.
8. **FAIR alignment as precondition. Do not retrofit FAIR.** The eight practices, adopted on day one, supply the four FAIR axes at low marginal cost. Adopted at submission time, the cost dominates: the manuscript has to be re-decomposed into claims, transcripts reconstructed if not preserved, rungs assigned in retrospect with the verification overhead now maximally expensive.

The integration is the contribution. None of the eight is new on its own.

5 Provenance Analysis

The graph at `doc/provenance.ttl` is not a passive record; it is a queryable structure whose health *is* the audit pass on **Reusable** (§3.1). Figure 7 renders the schema with the current instance counts read at compile time; the populated verification ladder is on the conference poster (`slides/poster-A0.tex`). The four diagnostic SPARQL queries the audit reads — rung distribution, section-to-claim coverage, source invocation strength, activity timeline — and the auto-generated tables they produce are walked in Appendix E.

The verification ladder rendered as a finite-state machine with live rung populations is on the conference poster (`slides/poster-A0.tex`); the queries that produce those populations are walked in Appendix E.

The four queries are diagnostic, not exhaustive. They surface the failure modes of §7 that have a graph signature — ghost citations, drift between prose and graph, sections without curated claims — and explicitly do not surface **semantic** drift (a claim correctly recorded in the graph whose prose has shifted away from its evidence). That is the readability scrutinizer's job (§3.2). A live interactive view of the graph is published at [the Provenance tab of the public site](#).

6 Process Evolution During Cooperative Writing

This section is a candid record of how the F(AI)²R pipeline itself changed while this paper was being written. The manuscript was produced under the same pipeline it documents (the fifth

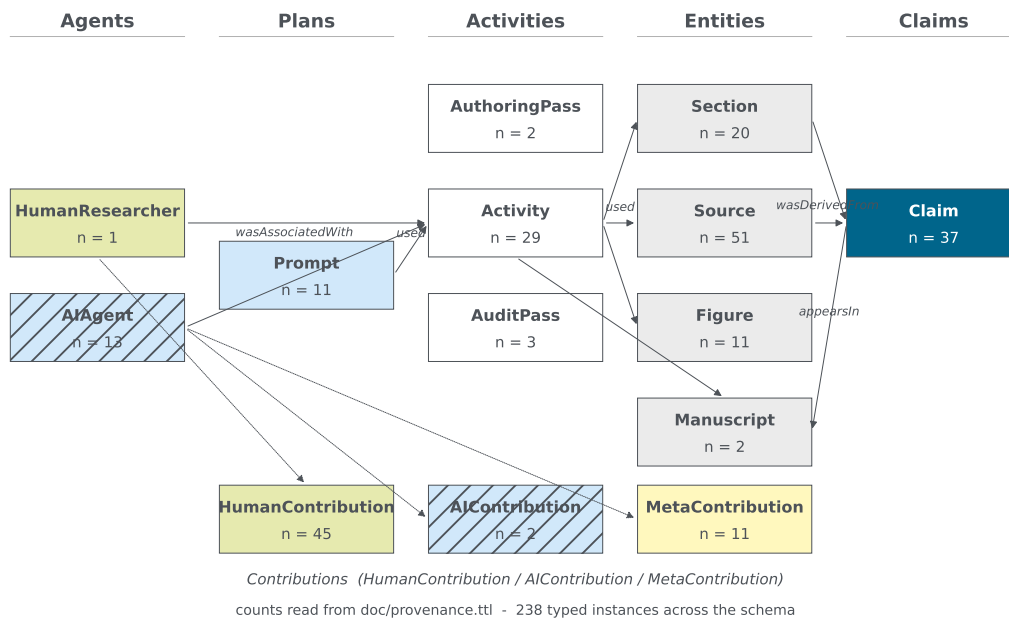


Figure 7. Shape of the F(AI)²R provenance graph, read from the live `doc/provenance.ttl` at compile time. Five PROV-O lanes (Agents, Plans, Activities, Entities, Claims) plus a Contribution satellite band; each node is decorated with its current instance count.

integrated practice, §4, item 5); the goal here is transparency about how the paper came to be and how researcher and AI interacted. The record is reconstructed from `doc/logbook.md` and `doc/provenance.ttl`.

Summary of evolutions. Table 1 synthesises the changes the cooperation imposed on its own scaffolding while the paper was being written. Five classes of evolution show up: *rules* (binding constraints added to one or more agent prompts), *agents* (a new specialised agent added to agents/), *schema* (a class or property added to `doc/provenance.ttl`), *pipeline* (a build, deployment, or rendering change), and *manuscript* (a structural reframing of the paper itself). The full chronology lives in `doc/logbook.md`; the table below is a deduplicated digest of the structurally-significant evolutions and is auditable against the forty-eight typed entries in `doc/user-contributions.md`.

Table 1. Process evolutions during cooperative writing. *Class*: rule (binding constraint), agent (new specialised prompt), schema (graph extension), pipeline (build / render / deploy change), manuscript (structural reframing). *Trigger / impact*: the researcher prompt or audit finding that landed the change and what it unlocked downstream. Reconstructed from `doc/logbook.md` and `doc/user-contributions.md`.

When	Class	What changed	Trigger / impact
2026-05-06	Rule	Primary-artefact consistency invariant (manuscript, graph, logbook, slides must agree).	Researcher direction; orchestrator named keystone; FAIR-aligner surfaces desync as audit fail.
2026-05-06	Rule	Chapter-per-file rule for <code>.tex</code> .	paper/main.tex refactored to a thin assembler; one <code>\section</code> per file; enabled chapter-level handback.

When	Class	What changed	Trigger / impact
2026-05-06	Rule	Page budget (~10 pp body) and short-form pivot.	Researcher framing as a position paper; condenser agent repurposed as page-budget enforcer.
2026-05-06	Rule	Source-research mandate; Consensus / Scholar default.	Cut hallucinated-citation surface; paywall-escalation sub-rule shipped together.
2026-05-06	Rule	Contribution-tracking rule (this file's existence).	Researcher question " <i>what is my contribution?</i> "; spawned <code>user-contributions.md</code> + <code>fair2r:Contribution</code> schema branch.
2026-05-06	Schema	<code>fair2r:Contribution</code> class with seven sub-types.	Operationalises the partition the Author's Note describes; mirrors per-side audit.
2026-05-06	Schema	<code>ai-confirmed</code> verification rung (between <code>lit-retrieved</code> and <code>lit-read</code>).	Felt cost of single-tier verification; rung added rather than weakening the existing two.
2026-05-06	Schema	<code>fair2r:Slidedeck</code> class; slides as fourth primary artefact.	Researcher request for two decks; presentation agent owns them.
2026-05-06	Agent	<code>presentation</code> agent added to <code>agents/</code> .	Slides became binding artefact; new prompt minted rather than overloading <code>scientific-writer</code> .
2026-05-06	Manuscript	Position-paper reframing (Position + Objections sections).	Researcher direction " <i>fair should be more a position paper</i> "; reshaped the contribution narrative.
2026-05-06	Manuscript	Author's Note carried over from <code>noheton/Obcurity-Is-Dead</code> .	Voice-and-stance anchor; the methodology's first-person register lives there <i>only</i> .
2026-05-06	Pipeline	Static site builder + Pages workflow + paper-build CI.	Made the artefact public from day one; cache-bust on the deployed site followed when staleness surfaced.
2026-05-06	Pipeline	Marp slide pipeline retired; Beamer-only.	Researcher direction " <i>only ship beamer</i> "; one toolchain to maintain instead of two.
2026-05-07	Rule	DLR Corporate Design pinned for every illustrator tool.	Default tool theming counts as an unfinished figure; written above the tool list, not below.
2026-05-07	Rule	Illustration toolset opened beyond TikZ (matplotlib, mermaid, networkx, graphviz, ...).	Researcher direction; <code>ladder-populations</code> and <code>contribution-histogram</code> matplotlib figures unlocked.
2026-05-07	Rule	No-parentless-claim rule (every claim must name a generating activity).	Eight-claim defect found by the verification example; rule landed by the curator pass to prevent recurrence.

When	Class	What changed	Trigger / impact
2026-05-07	Schema	<code>fair2r:repairs</code> object property.	Curator pass: per-claim repair activity carries the reason as <code>rdfs:comment</code> ; reasons are queryable.
2026-05-07	Agent	<code>research-protocol</code> agent added to <code>agents/</code> .	Standalone prompt for verification-axis research notes; produced <code>provenance-verification.md</code> .
2026-05-07	Pipeline	Pages source switched from branch <code>deploy</code> to GitHub Actions.	Default branch became <code>main</code> ; CI is single source of truth for the live site.
2026-05-07	Pipeline	LaTeX-fragment stripper in the site builder.	Markdown sources embedded <code>\emph{}</code> , <code>\textsc{}</code> , <code>\begin{enumerate}</code> ; the rendered Pages site needed a clean prose pass.

Chronology. The per-step chronology — which prompt triggered each row, what the change unlocked downstream, and why the cooperation imposed it on its own scaffolding rather than working around it — lives in Appendix D. Three threads from that chronology are load-bearing for the position taken in §2: *researcher-as-corrector* (the correction work is the human’s — catching wrong DOIs, drift between claim and evidence, attribution slips — and the volume of it across this paper’s history is itself an argument for the verification ladder, which gives every correction a canonical home as a state transition on a named claim); *pipeline-as-target-not-yet-as-practice* (the ten-stage pipeline of §3.2 is enforced on this manuscript by the discipline of the human author rather than by software, and the gap between architecture and automation is recorded honestly rather than papered over); and *logbook-as-conscience* (the append-only logbook is the index back into the provenance graph, and a session that produces no logbook entry produced no consequential work — by definition).

7 Failure Modes Named and Addressed

The pattern is meant to resist a finite set of failure modes that the literature on LLM-assisted writing has documented. Table 2 names them and the practice that bears each load. Three modes are out of scope and are stated as residual.

Residual. Three failure modes are out of scope. *Pre-publication leakage* — a model’s training data containing the manuscript before it is meant to — is a vendor problem F(AI)²R cannot reach. *Audience fatigue* — readers who have stopped engaging with disclosures because the boilerplate is identical at every venue — is a community problem the pattern can only signal toward. *Adversarial review* — a reviewer who reads the prose and ignores the graph, or vice versa — is a sociological problem that no artefact discipline can solve. We name these as residual.

8 Limits, dual-use, and standing objections

The pattern is not free. We owe the reader an explicit account of what F(AI)²R consumes, what it depends on, and what the standing objections to it look like.

Failure mode	Mitigation in the pattern
Fabricated citations	Verification ladder + per-claim provenance maps. A citation cannot reach LIT-RETRIEVED without a resolving identifier; cannot reach AI-CONFIRMED without a quoted snippet; cannot reach LIT-READ without a human reader. Reported base rates: 18–55% for full-citation generation [11], 17–34% for retrieval-augmented legal tools [12].
Claim–evidence drift	Handback discipline. The writer cannot rate its own claims; the Source Analyzer cannot rewrite prose; the Aligner files hand-backs.
Stoppification (drift in quoted passages)	Mirror discipline applied to quotations. Every quoted passage is either verbatim from a vendored source or a literal in the graph; a paraphrase that has crept into quotation marks fails the audit.
Model collapse	Indirectly mitigated. Transcript preservation plus disclosure lets downstream training corpora filter; the pattern does not prevent collapse but does provide the labelling [34, 35]
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List-of-lists prose	Readability scrutinizer flags occurrences in a defect registry; the writer rewrites.
Hedging chains	Readability scrutinizer flags hedge density above one per six to eight sentences; rule: each hedge paired with a commitment in the same sentence.
Placeholder dishonesty	Asset honesty over asset completeness. Placeholders must visibly identify themselves as AI-authored and pending.
Dual use (the pattern as a recipe for bad-faith authoring at scale)	Not eliminated. Made visible. The artefacts the pattern requires are themselves the surface that adversarial integrators inherit; a bad-faith adoption is detectable by inspection of those artefacts in a way that a bad-faith opaque process is not.

Table 2. Named failure modes and the integrated practice that bears the load. Eight modes; eight practices; the assignment is many-to-many in detail and one-to-one in load-bearing weight.

Resource cost and frontier-model dependence. Every authoring pass and every audit pass burns compute, and the verification ladder amplifies this: a contested claim is read by an agent, by a human, sometimes by a second agent during readability review, and again by the FAIR aligner. The energy and water footprint of frontier-model inference is non-trivial and is paid by someone, somewhere, regardless of who pays for the API call [36, 37, 38, 39]

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. We add the following request to the sixth integrated practice (§4, item 6): disclosure should *include the order-of-magnitude inference budget consumed by the manuscript*, declared at submission time alongside the rung distribution. Effective behaviour also depends on the underlying model and shifts as models change [40]

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; the transcripts of practice 1 and the version manifest in `doc/provenance.ttl` are partial mitigation, and any `fair2r:Activity` record should carry an explicit `fair2r:modelId` property

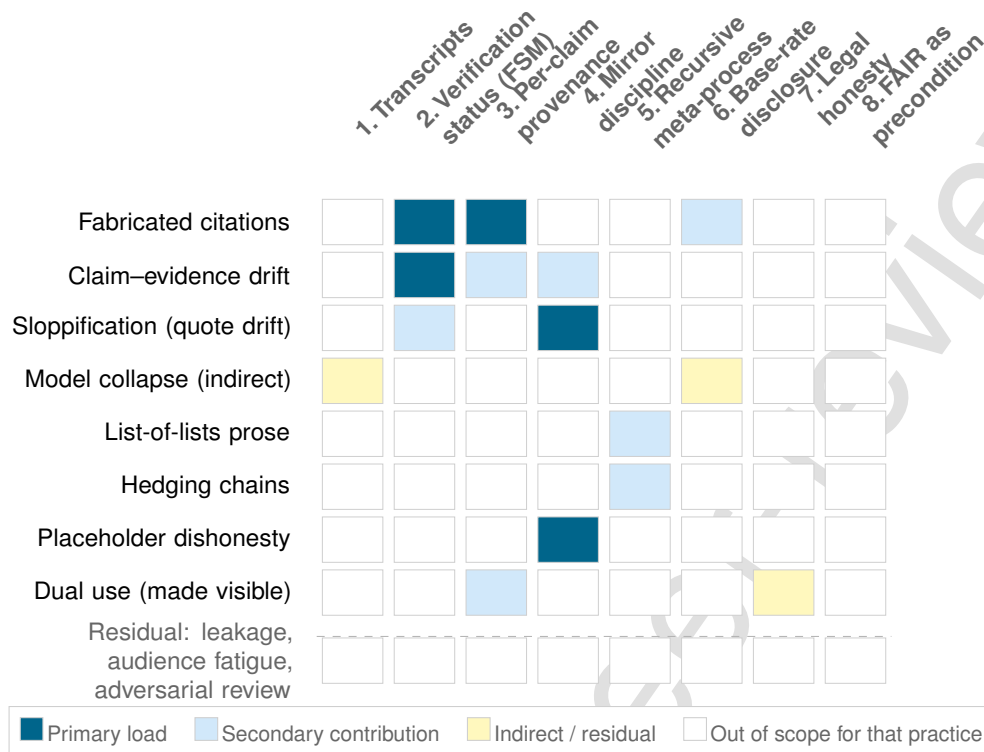


Figure 8. Coverage of the eight named failure modes by the eight integrated practices. A primary cell (solid blue) marks the practice that bears the failure-mode load; a secondary cell (soft blue) marks a contributing practice; an indirect/residual cell (yellow) records that the mode is mitigated only at the labelling layer. The bottom row records the residual failure modes the pattern can only signal toward (§7, *Residual*). The shape of the figure is the argument: no failure-mode row is empty, no practice column is unused, and the integration — not any single practice — is what produces full coverage.

pinned to a stable identifier so that a deprecated agent at minimum has a known headstone.

Equity. Research groups without access to frontier models cannot apply this pattern at the same effective rung; the verification ladder requires an AI agent at AI-CONFIRMED to actually understand the source it has read, and the floor for “understands” is set by the state of the art. F(AI)²R *describes* the work; it does not *equalise* access to it [41, 42]

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. We name this honestly and offer two partial mitigations: smaller-model fallbacks for routine audit passes (layout, hedging-density, list-of-lists detection) that do not need frontier-grade reasoning; and the structural property that an honestly-declared lower-rung distribution at submission time is preferable to a forged higher-rung distribution from a richer group, with the verification ladder making the difference legible to the reviewer.

Adjacent proposals and adjacent vocabularies. A concurrent maximalist proposal, the *Agent-Native Research Artifact* (Ara) of Liu et al. [43], replaces the narrative paper with a four-layer package — scientific logic, executable code with full specifications, an exploration graph of failed branches, and evidence grounding every claim — and reports reproduction-success and QA-accuracy gains on PaperBench and RE-Bench. F(AI)²R declines that bargain. We keep the human-readable manuscript as the primary artefact and ride a PROV-O graph alongside it; Ara’s exploration layer is a strict superset of our logbook plus claim-graph, and indicates a credible upper bound for what the same substrate could carry if a venue ever demanded it.

F(AI)²R is also sometimes mistaken for a dataset-metadata proposal. It is not. The dataset-metadata slot is already occupied by MLCommons Croissant 1.0 [44], a schema.org/JSON-LD vocabulary for ML datasets and their responsible-AI annotations. Croissant describes the *thing learned from*; F(AI)²R describes the *authoring process* that produces a paper. The two compose: a Croissant record can be referenced from a `fair2r:Source` triple, but neither schema subsumes the other.

Standing objections. A position paper that does not name the objections it expects has not committed to a position.

“This is just FAIR with extra steps.” Two prior re-readings (FAIR4RS [6], FAIR4ML [7]) judged that adding sub-principles for new artefact classes was preferable to absorbing them. We follow that precedent; the audit pass on **Reusable** is the load-bearing case — replaying a writing pipeline is materially different from replaying a build.

“Eight practices on day one is too expensive.” Eight is the *minimum*: the failure-modes table (Table 2) shows that dropping any one exposes a specific failure mode the others do not cover. The AI-CONFIRMED rung was introduced exactly to attack the verification-overhead bottleneck (§3.3).

“Provenance graphs do not prevent fraud.” Correct, and we never claim otherwise. The graph aims to make fraud *detectable by inspection*: a bad-faith adoption leaves a graph whose edges contradict the prose, whose claims reference non-existent sources, or whose `prov:wasGeneratedBy` activities have no transcript. Each of those is a checkable defect; the contrast is with unstructured AI-assisted writing, where the same fraud leaves no artefact a third party can point at.

“You cannot check the model’s reasoning, only its output.” Yes; we never claim otherwise. The discipline makes the *output’s relationship to the input* checkable, even when the model’s internals are not. That is what the verification ladder, the Source Analyzer hand-off, and the human-reader rung are for.

“A position paper should not also be a worked example.” The fifth integrated practice (§4, item 5) requires that a paper which prescribes a discipline have been written under it. The recursive case study is not an aesthetic choice; it is how the position can be checked.

The objections F(AI)²R cannot answer — pre-publication leakage, audience fatigue, adversarial review — sit in §7 as residual.

9 Conclusion

The FAIR Guiding Principles were drafted for data and have aged well. The artefacts that LLM-assisted scholarly writing produces are not data, not software, and not models; they need their own re-reading. F(AI)²R is that re-reading: the canonical FAIR axes with an (AI) factor multiplied through them, squared because each axis demands two passes over every artefact.

The contribution is not the name. The contribution is the integration of eight individually unoriginal practices — transcript preservation, verification-status labelling, per-claim provenance maps, mirror discipline, recursive meta-process documentation, base-rate-anchored disclosure, legal honesty about authorship, and FAIR alignment as precondition — enforced by a ten-stage agent pipeline whose prose-owning and audit-owning roles are strictly separated through a handback discipline. The integration is the contribution.

Submission conditions. We will submit F(AI)²R to the Research Data Alliance and to FORCE11 as a candidate community recommendation when two preconditions are met: (i) human-author consent for community-recommendation submission is recorded in `doc/publication-consent.m` and (ii) at least one external case study has adopted the pattern on a paper unrelated to either

the source repository or this one, and the resulting artefacts are public under a permissive licence. The conditions are the minimum we need to distinguish a single-author idiosyncrasy from a transferable pattern.

Call to action. We release this paper, its prompts, its provenance graph, its build pipeline, and its public site as the worked example, and we invite a second case study — by a different author, on a different paper, in a different domain — as the precondition for community-recommendation submission. The repository is open and the licences are permissive. A successor framework that argues with this one, replaces it, or tightens it is not a refutation; it is the point.

The longer arc. F(AI)²R is not the destination, only the entry ticket. The bioinformatics precedent (Appendix G) suggests where the trajectory leads: under the volume pressure that LLM-assisted scholarship is now imposing, future research success will depend less on the solitary craft of a single manuscript and more on three properties this schema is designed to support — provenance models that turn published claims into machine-readable hypotheses for the next study, graph interconnectivity across institutional knowledge graphs (a fully-instrumented Helmholtz cross-centre graph being the proximate example [45]), and clever experimental validation that uses the graph to identify the next testable deviation from the published consensus. The eight integrated practices serve all three. We are not arguing F(AI)²R has to be the standard; we are arguing that some standard with these properties is now load-bearing for the next decade of scholarship, and that naming the practice we are already drifting toward is the cheapest way to make it contestable in the open literature.

The bridge to what is now unthinkable. There is a generational analogue that names this responsibility clearly. Xennials — born late enough to grow up without the internet and early enough to spend their professional life inside it [46]

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— remember a working day in which a question that could not be asked of a search box was instead asked of a colleague, a reference librarian, or a card catalogue. The discipline that produced was not nostalgia but constraint: each step cost something, and the costs shaped the questions. A reader who has only ever known the present tooling cannot reconstruct that discipline by reading about it any more than a researcher who has only ever published in journals can reconstruct what preprint culture replaces. F(AI)²R is in part an attempt to encode it.

Senior researchers — those of us who remember why a methods section was structured the way it was, why a citation was load-bearing rather than decorative, why a draft circulated for a week before it left the office — carry a responsibility analogous to that bridge generation: to articulate, in machine-readable form, what was previously unthinkable to question, so that the practice that was load-bearing in our training survives translation into a toolset that no longer demands it. The eight integrated practices are the first crossing; the verification ladder, the PROV-O graph, and the handback discipline are its rails. We do not preserve them out of conservatism. We preserve them because the next cohort, raised inside a tooling whose default is fluency, deserves to know what made the discipline necessary before the choice to keep it or replace it is theirs.

The pipeline is the team; the team is the methodology. Specialisation reduces hallucination at the seam. Naming what we are already doing is the first step toward surrendering it to peer scrutiny.

Statement of authorship and AI use

This statement adapts the *Statement of independence* from the methodological ancestor [8] for a paper that is, by contrast, institutionally embedded.

Author. Florian Krebs (ORCID [0000-0001-6033-801X](https://orcid.org/0000-0001-6033-801X)) is the sole human author. The work is conducted at the German Aerospace Center (DLR / *Deutsches Zentrum für Luft- und Raumfahrt*), Zentrum für Leichtbauproduktionstechnologie (ZLP), Augsburg. The author is responsible for every claim in the manuscript and for the integrity of the artefacts the manuscript ships with.

Institutional context. DLR is a member of the Helmholtz-Gemeinschaft Deutscher Forschungszentren. Two Helmholtz initiatives are relevant to this work and are acknowledged here in addition to the author’s affiliation: the Helmholtz Metadata Collaboration (HMC) [45]

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, whose mandate is FAIR metadata across the Helmholtz association, and NFDI4Ing [47]

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, the DFG-funded research-data infrastructure for engineering sciences. F(AI)²R intersects both mandates and is offered to both communities for refinement.

AI use. The manuscript was drafted in cooperation with a Large Language Model (Anthropic Claude, model identifier `claude-opus-4-7`) operating under the per-role prompts committed to `agents/` and the harness rules in `CLAUDE.md`. The model is acknowledged here per the seventh integrated practice (§4, item 7); it is *not* an author. The legal position is consistent across the jurisdictions surveyed: in Germany (§2 *UrhG* [48]), in the United States (USCO 2023 [49]), and at the major medical journals via the ICMJE [33] and at *Science* via the editor-in-chief [17], AI cannot hold copyright, cannot consent to publication, and cannot be held accountable. The provenance graph in `doc/provenance.ttl` records, for each section, the activity that produced it, the agents (human and AI) that participated, and the verification rung of every named claim. The `CITATION.cff` lists humans only.

Data and code availability. The repository¹ contains the $\text{\LaTeX}$ sources, the bibliography, the agent prompts, the build pipeline, the static-site generator, the PROV-O graph, the logbook, and the redaction and consent policies. Code, prompts, and RDF are released under MIT; manuscript prose is released under CC-BY-4.0. The DLR design-system assets vendored under `site/static/dlr/` are subject to upstream licensing terms.

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¹<https://github.com/noheton/f-ai-r>

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A Verification Ladder Specification

The ladder is the finite-state machine of §3.3. This appendix supplies the per-rung specification used by the Source Analyzer (§3.2) and the Aligner.

Rung	Required artefact	Promotion condition	PRISMA / GRADE floor
UNVERIFIED-EXTERNAL	—	Claim recognised as needing a source.	—
NEEDS-RESEARCH	Search query in research-protocol has been identified.	A candidate URL or DOI has been identified.	—
LIT-RETRIEVED	URL/DOI that resolves	A Source Analyzer reads the source and confirms claim support, with quoted snippet.	PRISMA: Identification
AI-CONFIRMED	Quoted snippet in <code>sources.md</code> plus <code>fair2r:Source entity</code>	A human reads the source and confirms the inference from source to claim.	PRISMA: Screening; GRADE floor: LOW
LIT-READ	Vendored source in <code>doc/sources/</code> (when licensable) or human signature in logbook	—	PRISMA: Included; GRADE floor: MODERATE

Table 3. Verification ladder, with promotion condition and crosswalk floor. The crosswalk is the floor each rung permits, not a guarantee of the GRADE ceiling.

Retraction. A claim is retracted by adding a `prov:Invalidation` activity that names the original `fair2r:Claim` as `prov:invalidated`; the original triple is not deleted. A retracted claim

cannot be invoked by the manuscript and is rendered visibly retracted on the public site.

Per-rung budget. At submission time, the manuscript reports the fraction of claims at each rung. A typical first submission carries a single-digit percentage at LIT-RETRIEVED, a moderate fraction at AI-CONFIRMED, and the load-bearing remainder at LIT-READ. The ratio is itself a disclosure under the sixth practice (§4, item 6).

B Prompt Taxonomy

Every prompt under `agents/` follows the same five-section structure so that a different harness can ingest it with minimal adaptation.

Role. One paragraph: what the agent is, in plain language, with a one-sentence non-goal. The non-goal exists to refuse scope creep at read time.

Primary-artifact consistency (binding). The shared invariant of §4, item 4, restated with a tailored clause for the agent’s role: which of (manuscript, graph, logbook) the agent updates directly, which it submits for update by another agent, and what it must not finish without.

You do. The actions the agent is permitted to perform, in bullet form. Each bullet is small enough to be checked at review time.

You do not. The actions explicitly out of scope, in bullet form. The list is short and load-bearing: an agent that needs to be told “do not invent citations” is one prompt change away from doing so.

Output. The artefact the agent produces and where it lands. Outputs are typed: a draft is a diff, an audit is a defect registry, a graph update is a triple bundle, a section file is `paper/sections/<slug>.tex`.

Granularity rules.

- One agent per role; one role per file.
- One section per `.tex` file under `paper/sections/`; one `.tex` per section. Soft target: 100–400 lines per file. The `scientific-writer` prompt enforces this.
- One activity per stage in `doc/provenance.ttl`; an activity that runs more than once gets repeated entries with distinct `prov:startedAtTime`.

Refusal conditions. Each prompt names at least one refusal condition. Common ones: *do not invent citations*; *do not write to the graph if a submission lacks a corresponding manuscript change*; *do not deploy the public site without recorded publication consent*. Refusal is a feature, not a defect.

C F(AI)²R Cell Mapping

The original FAIR Guiding Principles enumerate fifteen sub-principles. F(AI)²R adds, for each, an authoring-pass cell and an audit-pass cell that name the artefact involved. The full list is in `doc/fair.md`; Table 4 is a condensed view.

The table is a contract. Adopting F(AI)²R amounts to producing an audit pass artefact for every cell in the right column at submission time, and to declaring the residual gaps explicitly when an artefact does not yet exist.

Sub-principle	Authoring pass (AI ₁)	Audit pass (AI ₂)
F1 (identifier)	Mint Zenodo DOI; assign fair2r:Claim IRIs.	Verify identifier resolves; verify IRIs are unique.
F2 (rich metadata)	Populate CFF / CodeMeta / Zenodo metadata.	Verify metadata surfaces agree.
F3 (metadata link)	Reference the data identifier from metadata.	Verify the link is bidirectional.
F4 (registered)	Deposit at Zenodo; minimum DataCite metadata.	Verify deposit visible from the registry.
A1 (retrievable)	Public Git repo; static site.	Verify clone-and-build succeeds on a clean machine.
A1.1 (open protocol)	HTTPS / Git.	—
A1.2 (auth where needed)	Document redaction policy and consent gate.	Verify redacted markers cover the documented taxonomy.
A2 (metadata persists)	Metadata in Git history; immutable Zenodo deposit.	Verify post-retraction metadata still reachable.
I1 (formal language)	Turtle for provenance; $\LaTeX$ for prose; CFF / JSON for metadata.	Validate Turtle; lint CFF.
I2 (FAIR vocab)	PROV-O + DCTERMS + FOAF + fair2r:.	Verify namespaces stable.
I3 (qualified refs)	prov:wasDerivedFrom, fair2r:contradicts, etc.	Verify graph has no dangling refs.
R1 (multiple attrs)	Record fair2r:VerificationState, prov:wasAttributedTo, etc.	Verify rung distribution at submission time.
R1.1 (clear licence)	MIT for code/RDF; CC-BY-4.0 for prose.	Verify SPDX expressions.
R1.2 (provenance)	The provenance graph itself.	Replay the graph; report drift.
R1.3 (community std)	Follow FAIR4RS / FAIR4ML where applicable.	Cross-check against the relevant standard at audit time.

Table 4. F(AI)²R cell mapping. Authoritative source: `doc/fair.md` in the public repository.

D Process evolution: the chronology

This appendix expands the digest in Table 1 (§6) into a per-step chronology. The body of the paper carries the table because the table is what a reader needs at submission time; the chronology below is the longer record, kept here so the table is auditable against narrative without expanding the body.

Bootstrap. Researcher request: initialise a paper-plus-pipeline repository with prompts under `agents/`, a logbook under `doc/`, and a single PROV-O graph that follows the creation of the paper, distilled from a prior repository (`noheton/Obcurity-Is-Dead`) with all domain content abstracted away. The first AI draft read F(AI)²R as “AI participates twice in the loop”. Reading the prior repository carefully revealed the researcher’s original framing: (AI) as a factor multiplied through every FAIR axis, squared because of the two passes per axis. The paper draft adopts the latter. Lesson: when the AI’s first-pass interpretation differs from the researcher’s prior writing, the prior writing wins.

Primary-artefact consistency invariant. After the bootstrap, the researcher asserted that the manuscript, the PROV-O graph, and the logbook are primary artefacts and must remain

consistent and up to date at all times. The AI added a binding “Primary-artefact consistency” block to every prompt; named the Orchestrator the ultimate custodian and the Provenance Curator the keystone; specified that the FAIR Aligner surfaces a desync as an audit *fail*. The rule was recorded as a `fair2r:Claim`, attributed to the researcher, and as a `prov:Activity` revising every affected prompt entity.

Static site, Pages workflow, paper-build CI. Researcher request: a public site in the spirit of the prior repository, plus continuous integration. The AI added a static-site generator (parsing `doc/provenance.ttl` via `rdflib`), the corresponding GitHub Pages workflow with a Turtle-validation gate, and a paper-build workflow.

Chapter granularity for the writer. Researcher request: *each .tex file should represent one chapter*. The Scientific Writer prompt gained a binding “File granularity” block: one `\section{}` per file, soft target 100–400 lines, filename slug matches the `\label{sec:...}`. `paper/main.tex` became a thin assembler.

Recursive attestation, by request. Researcher request: *add a section that this paper is generated using exactly this process*. A dedicated *self-reference* section (`paper/sections/self-reference.tex`) was added at the time, then retired in a later trim pass after the same attestation had been absorbed into the Author’s Note (§) and into practice 5 of §4; `prov:Invalidation` on the section is recorded in `doc/provenance.ttl`.

This section, by request. Researcher request: *document the evolution of the process during cooperative writing*. This section. The `fair2r:Claim` entries for it sit at AI-CONFIRMED until reviewed.

Short-form pivot. Researcher request: write only the short form (10 pages of body plus references and appendices) rather than maintaining a long/condensed split. The `main-condensed.tex` target was retired; the Condenser agent was repurposed as a *page-budget enforcer* that fires when the compiled PDF exceeds the configured budget; the Scientific Writer prompt gained a page-budget rule.

Institutional design system. Researcher request: integrate the DLR design system (Frutiger / Arial, blue/green/yellow chapter accents, square corners, hairline borders, photography-led, no marketing register). The site CSS was rewritten on top of the DLR `colors_and_type.css` tokens; a small $\text{\LaTeX}$ style file (`paper/style/fair2r.sty`) carries the equivalent typography and rule colours into the PDF; the imprint was updated with the author’s institutional identity (DLR ZLP Augsburg, Helmholtz, NFDI4Ing, HMC).

Patterns observed

Three patterns recur and are worth naming.

Researcher-as-corrector. The researcher’s interventions are short, specific, and rule-shaped. None is a rewrite of AI-produced prose. The cooperation lives in the gap between a model that produces a plausible default and a researcher who can recognise when the default is wrong.

Pipeline as target, not yet as practice. The ten-stage pipeline is the architecture the prompts encode. The present draft was produced by a single AI agent in conversation with the researcher, with role separation documented in the prompts but not yet actively switched per turn. We name this gap rather than paper over it.

Logbook as conscience. Every event above corresponds to an entry in `doc/logbook.md`. The append-only discipline made the reconstruction cheap. *The logbook is the conscience of the cooperation; the provenance graph is its record.*

Field notes from the second cooperative writing project

The methodological ancestor [8] was the author's first cooperative-writing project. The present paper is the second. Three observations from the second-time experience are worth recording; the full set lives in `doc/user-observations-log.md`.

The researcher's role is shifting toward experimentation. Cooperative writing with an LLM, as practised here, lifts the prose-production load off the researcher and leaves more cycles for abstraction, decision-making, and design of the next experiment. For empirical disciplines this is a net good: writing-as-administrative-overhead shrinks; experimentation-as-the-actual-contribution grows. For paper-driven disciplines — philosophy, parts of the humanities — the picture is more uncomfortable: the prose *is* the experiment, and removing the slow, deliberate craft of writing risks removing the thinking that the writing was doing [50]

verify

. We name the asymmetry rather than pretend F(AI)²R is discipline-neutral.

Prompt-craft is closer to programming than to drafting. A second-time observation: writing a prompt feels easier than writing the corresponding paragraph oneself. The short conversational cycle maps, for a researcher trained in computer science, to the familiar loop of writing a program and watching its output — closer to experimentation than to formal academic writing [51, 52]

verify

. This is a feature (it lowers the activation energy of getting started, which is itself a contribution to throughput) and a risk (prompt-craft can become a comfortable substitute for the harder parts of writing). The risk is managed, in F(AI)²R, by keeping the human in the deciding seat (§, *What surprised me about the cooperation*) and by the per-claim provenance map that makes drift detectable.

Responsibility includes the rooms after publication. The model-side contribution ends at the commit. The human-side contribution continues into the rooms where the work has to be defended in person — conferences, panels, discussions, reviewer correspondence. AI cannot share that load: the model cannot consent, cannot carry a reputation, and cannot be cross-examined at the next RDA plenary or HMC symposium. The seventh integrated practice (§4, item 7) names the legal version of this point; the field-notes version is more ordinary. *Whoever puts their name on the paper signs up for every conversation the paper produces afterwards*, and the AI's contribution is, by construction, absent from those conversations. The interpersonal-sharing leg of authorship is the leg the methodology section cannot eliminate even in principle.

E Provenance analysis: queries and worked numbers

The body section (§5) summarises the audit; this appendix walks the four diagnostic queries on the live graph. The auto-generated tables below are produced by `scripts/provenance_analysis.py`, which the build pipeline re-runs on every commit. They reflect the state of `doc/provenance.ttl` at the time this PDF was compiled and are embedded into the prose by the build, not transcribed by hand.

Verification rung distribution. Every claim in the graph carries a `fair2r:verificationState`; we report the fraction at each rung at submission time per the sixth integrated practice (§4, item 6). The query is one SPARQL pattern:

```
SELECT ?state (COUNT(?c) AS ?n)
WHERE { ?c a fair2r:Claim ; fair2r:verificationState ?state }
GROUP BY ?state
```

The result for the present manuscript:

Rung	Claims	Share
needs-research	2	4.7 %
ai-confirmed	7	16.3 %
human-confirmed	24	55.8 %
source-vendored	9	20.9 %
human-read	1	2.3 %
Total	43	

Section-to-claim coverage. Every section in `paper/main.tex` should appear in the graph as a `fair2r:Section` entity, with at least one `prov:Activity` that produced it and (almost always) at least one `fair2r:Claim` attributed to that activity. A section that exists in prose but not in the graph is a defect surfaced by the FAIR Aligner.

Section heading	Label	In graph?	Claims
Process evolution: the chronology	<code>sec:appendix-evolution</code>	no	0
Provenance analysis: queries and worked numbers	<code>sec:appendix-provenance-analysis</code>	yes	0
Implementation: layout, CI, replay	<code>sec:appendix-implementation</code>	yes	0
Pattern extensions: domain ontologies and graph-as-research-infrastructure	<code>sec:appendix-pattern-extensions</code>	yes	0
Author's Note — Advice for reading	<code>sec:authors-note</code>	yes	6
Conclusion	<code>sec:conclusion</code>	yes	11
Eight Integrated Practices	<code>sec:eight</code>	yes	10
Process Evolution During Cooperative Writing	<code>sec:evolution</code>	yes	10
Failure Modes Named and Addressed	<code>sec:failure-modes</code>	yes	10
Introduction	<code>sec:intro</code>	yes	3
Limits, dual-use, and standing objections	<code>sec:limits-and-objectives</code>	yes	0
The F(AI) ²	<code>sec:pattern</code>	yes	10
Position	<code>sec:position</code>	yes	6
Provenance Analysis	<code>sec:provenance-analysis</code>	yes	2
Statement of authorship and AI use	<code>sec:authorship-statement</code>	yes	6

A coverage row that reads *prose: yes / graph: no* is a hard fail; *graph: yes / claims: 0* is a soft warning meaning the section has been registered but the claim curation has not caught up.

Source invocation strength. Every `fair2r:Source` entity that the manuscript cites should appear as the object of at least one `prov:wasDerivedFrom` edge from a `fair2r:Claim`. A source cited but never linked from a claim is most often a **ghost citation** — a reference that lives in the bibliography because the prose mentions it, but that no claim actually relies on. In a healthy F(AI)²R manuscript, the count is zero.

Sources cited: 56; invoked by at least one claim: 28; **ghost citations: 28.**

Source	Title	Invocations
aclrr_llm_policy	aclrr_llm_policy	1
alemohammad2023mad	alemohammad2023mad	0
anderson2024homogenization	anderson2024homogenization	0
ashburner2000go	ashburner2000go	2
bender2021stochastic	bender2021stochastic	0
benjelloun2024croissant	benjelloun2024croissant	1
birhane2022values	birhane2022values	0
chen2023drift	chen2023drift	0
chuehong2022fair4rs	chuehong2022fair4rs	0
clark1998extended	clark1998extended	0
clark2025extending	clark2025extending	0
clarke2009modelchecking	clarke2009modelchecking	1
conroy2023sleuths	conroy2023sleuths	1
curdt2025hmc	curdt2025hmc	1
eisen2018preprints	eisen2018preprints	1
else2023chatgpt	else2023chatgpt	1
gebru2021datasheets	gebru2021datasheets	0
guyatt2008grade	guyatt2008grade	0
hutchins1995cognition	hutchins1995cognition	0
iclr_llm_policy	iclr_llm_policy	1
icmje2023	icmje2023	1
ioannidis2005	ioannidis2005	1
janowicz2019sosa	janowicz2019sosa	1
klein2009sel4	klein2009sel4	1
kobak2024delving	kobak2024delving	1
krebs2026obscurity	krebs2026obscurity	10
kuteeva2024diversity	kuteeva2024diversity	0
li2023thirsty	li2023thirsty	0
liang2024mapping	liang2024mapping	1
liu2023prompt	liu2023prompt	0
liu2026ara	liu2026ara	1
luccioni2024power	luccioni2024power	0
maresh2024legal	maresh2024legal	2
mitchell2019modelcards	mitchell2019modelcards	0
neurips_llm_policy	neurips_llm_policy	1
page2021prisma	page2021prisma	0
patterson2021carbon	patterson2021carbon	0
pineau2021reproducibility	pineau2021reproducibility	1
qudt	qudt	1
ravi2024fair4ml	ravi2024fair4ml	0
reynolds2021prompt	reynolds2021prompt	0
rijgersberg2013om	rijgersberg2013om	1
sadasivan2023reliably	sadasivan2023reliably	0
schmitt2020nfdi4ing	schmitt2020nfdi4ing	0
shafer2014xennials	shafer2014xennials	1
shumailov2024collapse	shumailov2024collapse	0
strubell2019energy	strubell2019energy	0
tennant2016open	tennant2016open	1

Activity timeline. The `prov:Activity` nodes form a dated record of every substantive change. The most recent five activities, in ISO-8601 order, are:

Date	Activity	Label
2026-05-13	integrate-external-works	Integrate five external works (Ara, Vaswani, Croissant) into the manuscript
2026-05-08	provenance-graph-hygiene	Provenance graph hygiene pass after audit
2026-05-08	fix-poster-density	Fix portrait poster vertical density; add density rule to layout-scrutinizer
2026-05-08	expand-layout-scrutinizer	Expand layout-scrutinizer scope to all five primary artefacts; flip poster from
2026-05-07	write-provenance-verification	Author the provenance-verification scoping note

The same data, with one row per session, is the human-readable mirror in `doc/logbook.md`; drift between the two is itself a defect (see §3.4 on the handback discipline).

Keep this appendix honest. This appendix is generated against the current graph. If a future revision adds a verification rung, a new section, or a new failure mode, the corresponding entry here updates automatically on the next build; the prose around the figures stays put. The intent is that a reader who clones the repository, recompiles the PDF, and re-reads this appendix, will see the analysis reflect the manuscript that was actually produced — not a snapshot frozen at draft time.

F Implementation: layout, CI, replay

The pattern is materialised in the public repository that produced this paper.² The intent is not to prescribe a stack: any version-control system with a build target, a structured-data format that supports PROV-O, and a way to version prompt files would do.

Layout. `paper/` carries the $\text{\LaTeX}$ sources, bibliography, Makefile, and figure sources, with one `\section{}` per file under `paper/sections/`. `doc/` carries the methodology, FAIR notes, human-AI collaboration policy, research protocol, submission plan, the append-only logbook, the PROV-O graph in Turtle, and a Mermaid mirror. `agents/` carries one markdown file per role (orchestrator, scientific writer, source analyzer, FAIR aligner, provenance curator, layout scrutinizer, readability reviewer, illustration, condenser, research protocol), each structured per the prompt taxonomy of Appendix B. `scripts/` carries the static-site generator that parses `doc/provenance.ttl` via `rdflib` and renders it into a small HTML site.

Continuous integration. `build-paper.yml` compiles the manuscript on every push that touches `paper/` and uploads the PDF as an artefact, using `xu-cheng/latex-action@v3`. `pages.yml` validates the Turtle, builds the site, and (on `main`) deploys via the official GitHub Pages action chain. Working branches see the build output as a downloadable artefact, never as a public deploy.

Replay loop. A reader who clones the repository, installs a TeX Live distribution and the dependencies in `scripts/requirements.txt`, runs `make -C paper pdf` and `python scripts/build_provenance_si`

²<https://github.com/noheton/f-ai-r>

has reproduced the paper, the provenance graph, and the public site. The audit pass for **Reusable** is, in our setup, that loop run on a clean machine, compared against the published artefact. Any difference is a defect.

G Pattern extensions: domain ontologies and graph-as-research-infrastructure

This appendix expands two forward-looking subsections that the body of §3 points at: how the pattern composes with domain-specific ontologies, and how the bioinformatics precedent for semantic-tagging-as-norm reframes provenance graphs as research infrastructure under volume pressure. Neither is on the critical path of the eight integrated practices, but each is how F(AI)²R generalises beyond the present case.

G.1 Extensibility through domain ontologies

PROV-O is the spine; it does not have to be the only ontology in the graph. The schema we have introduced is compositional: any `fair2r:Claim` can carry additional triples that ground its *values* in a domain ontology, alongside the `prov:wasDerivedFrom` edge that grounds its *evidence*. The two axes are orthogonal — provenance asks *where did this claim come from?*, domain ontologies ask *what does the value of this claim mean?* — and a manuscript that answers both is more reproducible than one that answers only the first.

A neighbouring vocabulary worth naming explicitly is MLCommons Croissant 1.0 [44], the schema.org / JSON-LD vocabulary for ML datasets and their responsible-AI annotations. Croissant occupies the dataset-metadata slot (§8) and composes cleanly with the F(AI)²R schema: a `fair2r:Source` can reference a Croissant record so that the provenance graph for a manuscript and the dataset metadata for the data it sits on top of read as a single linked-data graph rather than two disconnected ones.

Two examples concrete enough to plug in today:

Sensor data: SOSA / SSN. The W3C Semantic Sensor Network ontology and its lightweight core SOSA [53]

verify

expose `sosa:Observation`, `sosa:Sensor`, `sosa:FeatureOfInterest`, and the relations between them. A `fair2r:Claim` that asserts a measurement can carry a `prov:wasDerivedFrom` pointing at the underlying `sosa:Observation`, and the audit pass gains the ability to ask *is this measurement well-formed?* as a SHACL query rather than a paragraph of prose.

Units of measure: OM-2 / QUDT. The OM-2 ontology of units of measure [54]

verify

and the QUDT vocabulary [55]

verify

expose `om:Quantity`, `om:Unit`, `qudt:hasUnit`, and the dimensional relations between units. A claim that asserts a numeric value can carry the unit as an explicit triple rather than as a typographic decoration. Unit consistency is then a graph query, not a manuscript-reading exercise.

The class of failure modes this addresses is the one prose alone cannot reach: a value that is internally coherent in its prose and its citation but *wrong in its units*, or *wrong in the dimensional sense it implies*. A reviewer catching such a defect today reads the figure caption and the

methods section and triangulates; an audit pass with a domain ontology reads the graph and the SHACL shapes and reports.

The recommendation is open. F(AI)²R does not prescribe specific domain ontologies because the right choice is discipline-specific. It prescribes that, where a discipline already has a *konkrete standardisierte Methode* expressed as an ontology, the F(AI)²R repository should adopt it and link from `fair2r:Claim` entities to its terms. NFDI4Ing and the Helmholtz Metadata Collaboration are the nearest two communities to this paper that maintain such ontologies for engineering and metadata respectively, and the integration cost is low. The ceiling is the discipline's; the floor is what F(AI)²R asks for.

G.2 Looking forward: provenance graphs as research infrastructure

The argument of §G.1 generalises. The discipline that pioneered semantic-tagging-as-norm under data-volume pressure is bioinformatics: faced with what was, at the time, an obscene amount of sequence data, the field standardised identifier and annotation schemas (the Gene Ontology [56]

verify

being the most visible example) so that genes, proteins, and their relations could be referenced unambiguously across labs, papers, and databases. The ontology was not a research output; it was a precondition for research that scaled. The community's investment in semantic tagging was, in retrospect, what made the next decade of bioinformatics possible.

We claim a closer analogy than is sometimes recognised. LLM-assisted scholarly writing is producing artefacts at a volume that the unaided reviewer pipeline cannot absorb (§1). The response will, on present trajectory, look more like the bioinformatics response than like the editorial response: not new detection tools but new shared schemas, ontologies, and graph-interconnected metadata that let claims, sources, and activities reference each other across institutional boundaries. **Future research success may come to depend less on the solitary craft of a single manuscript and more on three things the F(AI)²R schema is designed to support:** (i) provenance models that turn published claims into machine-readable hypotheses for the next study; (ii) graph interconnectivity across institutional knowledge graphs — the Helmholtz cross-centre graph [45] is the proximate example, and a fully-instrumented version would let a claim in one centre's manuscript be linked, queried, and challenged by claims in another's without leaving the linked-data layer; (iii) clever experimental validation that uses the graph to identify the next testable deviation from the published consensus.

That trajectory is a position the paper is willing to take and a direction the eight integrated practices serve. The verification ladder feeds (i); the per-claim provenance map and the domain-ontology side-arms feed (ii); the recursive case study and the audit pass feed (iii). None of this requires F(AI)²R to be the standard. It requires only that some standard with these properties be adopted, and the present paper is one offer among several the community can refine.